International Institute of Information Technology, Hyderabad
CS9.312-Introduction to Quantum Information and Computation (Spring-2026)

Quiz

Due: 7 Feb, 2026 Total Marks: 25

Question 1 [13 Marks]. Suppose $x \in \{0,1\}$ is a single bit. Then the state $|x\rangle$ denotes either $|0\rangle$ or $|1\rangle$. Recall that $|+\rangle = \frac{1}{\sqrt{2}}\sum_{x\in\{0,1\}}|x\rangle$. Moreover, the Hadamard gate satisfies $H|0\rangle = |+\rangle = \frac{1}{\sqrt{2}}\sum_{z\in\{0,1\}}(-1)^{0.z}|z\rangle$ and $H|1\rangle = |-\rangle = \frac{1}{\sqrt{2}}\sum_{z\in\{0,1\}}(-1)^{1.z}|z\rangle$. Now consider a system of n qubits. Each computational basis state can be written as $|\mathbf{x}\rangle = |x_1\rangle \otimes \cdots \otimes |x_n\rangle$, where $\mathbf{x} = x_1x_2\cdots x_n \in \{0,1\}^n$ is an n-bit string.

(a) **[4 Marks]** Show that

$$H^{\otimes n}|0^n\rangle = \frac{1}{2^{n/2}}\sum_{\mathbf{x}\in\{0,1\}^n}|\mathbf{x}\rangle.$$

(b) **[6 Marks]** More generally, let $\mathbf{x} \in \{0,1\}^n$ be an arbitrary n-bit string. By applying the Hadamard gate to each qubit separately, show that

$$H^{\otimes n}|\mathbf{x}\rangle = \frac{1}{2^{n/2}}\sum_{\mathbf{z}\in\{0,1\}^n}(-1)^{\mathbf{x}\cdot\mathbf{z}}|\mathbf{z}\rangle,$$

where $\mathbf{x}\cdot\mathbf{z} = x_1z_1 + x_2z_2 + \cdots + x_nz_n \pmod 2$.

(c) **[4 Marks]** Let $f : \{0,1\}^n \to \{0,1\}$ be a Boolean function and let $|\psi\rangle = \frac{1}{2^{n/2}}\sum_{\mathbf{x}\in\{0,1\}^n}(-1)^{f(\mathbf{x})}|\mathbf{x}\rangle$. If $H^{\otimes n}|\psi\rangle = \sum_{\mathbf{y}\in\{0,1\}^n}\alpha_{\mathbf{y}}|\mathbf{y}\rangle$, then prove

$$\alpha_{\mathbf{y}} = \frac{1}{2^n}\sum_{\mathbf{x}\in\{0,1\}^n}(-1)^{f(\mathbf{x})}(-1)^{\mathbf{x}\cdot\mathbf{y}}.$$

Question 2 [12 Marks]. Let us consider following matrices.

$$U_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix}, \quad U_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \text{ and } U_3 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 0 & 0 & i \\ 0 & 1 & i & 0 \\ 0 & i & 1 & 0 \\ i & 0 & 0 & 1 \end{pmatrix}.$$

(a) **[3 Marks]** Prove that U_1, U_2, and U_3 are unitary matrices.

(b) **[9 Marks]** Determine which of U_1, U_2, U_3 are product and which of U_1, U_2, U_3 are entangling.